

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_005832 | Base: main-v1.0

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Decision: NO_GO | Confidence: 53%

CHANGE REQUEST

The structure must be longer downward by 40% to accommodate a larger battery pack.

CASCADE ISSUES IDENTIFIED

[CRITICAL] ISSUE-001 — Thermal Gradient Exceeds Limit

Subsystem: thermal | Req: L1-THM-001 | Type: constraint_violation

Increasing the structure height by 40% downward will likely increase thermal gradients across the optical bench, potentially exceeding the $\Delta T \leq 15^{\circ}\text{C}$ requirement.

[MAJOR] ISSUE-002 — Shock Test Required for New Geometry

Subsystem: mechanical | Req: L1-MEC-003 | Type: missing_test

The change in structure geometry necessitates a new shock test to ensure compliance with the launcher SRS ICD.

[MINOR] ISSUE-003 — Volume Envelope Exceeded

Subsystem: structure | Req: None | Type: constraint_violation

The increase in structure height may cause the payload volume to exceed the platform accommodation envelope of 0.10 m³.

MISSING ECSS TESTS — NOW REQUIRED

ECSS-E-ST-32C Rev.1 §9 — Shock test per launcher SRS ICD

Reason: The change in structure geometry requires revalidation of shock response.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	26%	FAIL	Thermal Gradient Exceeds Limit	■ MODIFIED
mechanical	68%	WARN	Shock Test Required for New Geometry	■ MODIFIED
structure	83%	OK	Volume Envelope Exceeded	■ MODIFIED
solar	89%	OK	Power margin +18%	—
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

- L1-THM-001 — verification evidence invalidated, re-analysis required
- L1-MEC-003 — verification evidence invalidated, re-analysis required